

8 D

$$T + mg = \frac{mv^2}{r}$$

$$1.2 + 0.040 \times 9.81 = \frac{mv^2}{0.30}$$

$$\frac{1}{2}mv^2 = 0.23886$$

By conservation of energy,

$$\frac{1}{2}mv^2 + mgh = \frac{1}{2}mu^2$$

$$0.23886 + 0.040 \times 9.81 \times (2 \times 0.30) = \frac{1}{2} \times 0.040 \times u^2$$

$$u^2 = 23.715$$

$$a = \frac{u^2}{r} = \frac{23.715}{0.30} = 79.1 \text{ m s}^{-2}$$

9 B Graph C, for path from S to Q means that the two stars have different masses (since there